PARTH PALA

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PROFESSIONAL SUMMARY

Lead Software Engineer with **8+ years of experience** in designing, developing, and delivering Android applications across Mobile, Android TV, Google TV, and enterprise mobility solutions. Successfully designed and contributed to **15+ Android applications,** including multiple applications achieving **100K+ downloads.** Worked with global clients including **Google**. Proven experience in architecting applications from scratch, leading technical discussions with clients, mentoring development teams, and driving end-to-end delivery of scalable mobile solutions. Strong expertise in Kotlin, Java, MVVM, Coroutines, Flow, gRPC, Google ML Kit, ExoPlayer, and modern Android development practices.

TECHNICAL SKILLS

- **Languages & Frameworks:** Kotlin, Java, XML, Android SDK
- **Architecture & Core Patterns:** MVVM, MVP, Clean Architecture, Repository Pattern, Coroutines, Flow, LiveData, Data Binding.
- **Networking & Communication:** Retrofit, REST APIs, Volley, gRPC, Ktor, SignalR, OkHttpURLConnection
- **Database :** SQLite, Room Database, SQL Server, SQL Server Management Studio (SSMS).
- **Testing:** Espresso, JUnit, Mockito.
- **Security :** OWASP, HTTPS, ProGuard, App-level Encryption & Decryption, local certificate pinning, EncryptedSharedPreferences.
- **Libraries & SDKs :** Google ML Kit, Google APIs(Crashlytics, Maps, Push), Facebook SDK, YouTube SDK, ZXing(Barcode & QR), Glide/Picasso, CameraX, ExoPlayer, Payment Gateway Integration.
- **OTHERS :** Accessibility (TalkBack), Deep Linking, Localization, QR/Barcode Integration, IIS & Kestrel deployment.

PROFESSIONAL EXPERIENCE

***NEC SOFTWARE SOLUTIONS , Mumbai* — Lead Software Engineer | August '25 – Present**
(Contract via StrideFuture Technologies: Aug 2025 – Dec 2025 | Converted to Full-Time due to performance: Dec 2025 – Present)

- Served as the sole Android Engineer, owning the complete Android application lifecycle from architecture and development to production deployment and ongoing enhancements.
- Defined the complete Android architecture, selected the technology stack, established coding standards, and developed reusable components to ensure a scalable and maintainable solution.
- Led technical discussions directly with client stakeholders, independently handling requirement gathering, solution design, sprint planning, effort estimation, sprint reviews, and technical demonstrations.
- Coordinated technical delivery with 2 Backend Developers and 1 Database Developer, providing guidance on API design, business logic, database changes, and sprint execution.

- Developed and optimized a real-time multi-face detection pipeline using Google ML Kit, achieving 15–20 FPS while supporting simultaneous face detection.
- Designed and integrated multiple communication mechanisms including SignalR, embedded Ktor Web Server, and Long Polling to support reliable backend communication under varying network conditions.
- Extended responsibilities beyond Android development by deploying backend services using IIS and Kestrel, managing SQL Server configuration through SSMS, and supporting production deployments.
- Received client appreciation for technical leadership, ownership, calm decision-making, and consistently delivering high-quality solutions within committed timelines.

INFOSYS, Mumbai — *Technology Analyst (Client: Google)* | *April '23 – August '25*

- Led a team of 5 developers and won an award for Employee of the Quarter for Google TV and related settings applications across global TV platforms.
- Regularly supported team members by actively debugging, unblocking tasks, and accelerating delivery.
- Resolved critical blocker nearing the SLO deadline, ensuring on-time release.
- Integrated high-performance gRPC protocols for efficient, low-latency API communication across internal Google services.
- Solved 30+ production bugs, including critical crashes and UI defects, contributing to stability improvements.
- Redesigned 4 major screens in Google TV Settings for a cleaner, accessible user experience and spearheaded 5+ feature rollouts including remote pairing, login, and localization setup.
- Enhanced accessibility across Google TV platform through TalkBack support and content descriptions.
- Diagnosed and resolved OS-level bugs, including 10+ Priority issues, significantly reducing platform escalations and improving reliability.
- Utilized Google tools for bug tracking, code reviews, and project workflows, and worked across devices like Google Chromecast, Android/Google TVs for testing and validation.

Senior System Engineer* - *July '21 - April '23.

- Increased unit test coverage from approximately 45% to 89% by restructuring components and improving application testability.
- Improved application stability through performance optimization, code refactoring, and enhanced barcode scanning implementation.
- Collaborated closely with cross-functional teams to deliver scalable Android solutions following Agile development practices.

MOBITRAIL, Mumbai — *Android Developer / Sr. Software Programmer | Sept '17 – July '21.*

- Developed and maintained enterprise Android applications using Java, Kotlin, MVP, and MVVM architectures.
- Successfully published 4 Android applications on Google Play Store, achieving more than 100,000 downloads.

- Improved application stability by reducing crashes through performance optimization, modular architecture, and proactive bug resolution.
- Collaborated closely with QA, UX, Product Managers, and clients to ensure high-quality releases across multiple Android devices.
- Implemented secure Android applications following OWASP guidelines, encryption techniques, and industry-standard development practices.

Key Projects

• US Border Security Application (NEC)

Description: Android application for secure device verification and real-time surveillance workflows.

- Architected and developed the complete Android application from scratch, including application architecture, reusable components, and technology selection.
- Designed a resilient communication mechanism using SignalR, embedded Ktor Web Server, and Long Polling to ensure reliable communication with backend services.
- Integrated ExoPlayer for live video streaming and optimized Google ML Kit face detection to support real-time processing of multiple faces.
- Built configurable runtime controls enabling dynamic operational settings without requiring application redeployment.

• Google TV & Google TV Settings

Description: Android TV platform and Google TV system applications used across consumer smart TV devices worldwide.

- Delivered 10+ high-impact features, including remote pairing, language detection, and GMS setup flows.
- Led accessibility overhaul for TalkBack, improving navigability and readability across Google TV applications.
- Fixed 30+ production issues (crashes, UI bugs); redesigned 4 critical screens in Settings app.
- Resolved 10+ P1 bugs through critical OS-level debugging and low-memory UI optimizations, significantly improving platform stability and reducing escalations.

• BPOST MY ROUND

Description: Parcel management app for Belgian postal workers.

- Addressed key blockers in unit testing, raising code coverage from ~50% to 89%, and improved barcode scanning accuracy and overall app responsiveness.

• ICICI Face-Based Attendance App

Description: Application for real-time attendance marking using CameraX and ML Kit.

- Implemented CameraX + ML Kit face detection (contours, eye-blink, tilt, head orientation) with Text-to-Speech (TTS), Speech-to-Text (STT), and real-time overlay UI for interactive attendance confirmation.

• **Experian – Your Choice (YRMS)**

Description: Internal HR app for Experian employees across 15+ countries with 1k+ downloads.

- Designed, developed 12–15 modules (salary, rewards, support) from scratch in 25 days, praised by client for fast, stable rollout using encryption protocol; led full-cycle ownership of the app.

• **IIFL INVESTORQ**

Description: Finance-focused Q&A app with 100k+ downloads.

- Designed and developed 20+ modules from scratch; led full-cycle ownership of the app.
- Took ownership of the application when stability declined to 60–70% crash-free, improving it to 99–100% through performance optimization and proactive issue resolution.
- Implemented encryption, error handling, and OWASP security standards.

Open Source & Personal Projects

- **Stoppy:** Customizable stopwatch application featuring real-time theme, font, and lap color personalization with persistent state management.
- **WarrantyMate:** Warranty management application that securely stores product warranty details and provides timely expiry reminders to help users track warranties efficiently
- **FourInOneViews:** Open-source Android UI library that combines four independent views into a single reusable component, reducing layout complexity, improving maintainability, and minimizing UI rendering overhead.
- **Splitter:** Expense-sharing application that calculates individual contributions by item, automatically generating accurate per-person expense summaries.
- **FFF's:** Two-player real-time quiz game featuring a split-screen interface where players compete simultaneously, with automated score tracking and winner determination.

EDUCATION: BACHELOR'S OF SCIENCE IN INFORMATION TECHNOLOGY (BSC-IT), MUMBAI 2017.